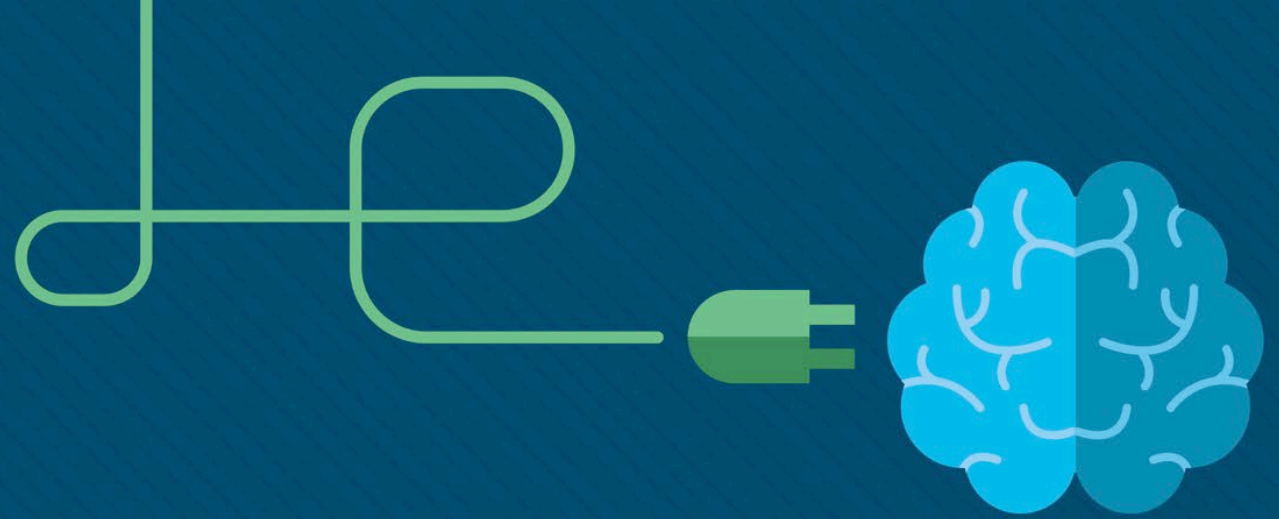




Week 6

RH124 Chapter 9. Control Services and Daemons

RH124 Chapter 11. Managing Network



1 The *systemd* Daemon

- The *systemd* daemon manages the startup process for Linux, including service startup and service management in general. The *systemd* daemon activates system resources, server daemons, and other processes, both at boot time and on a running system.
- *Daemons* are processes that either wait or run in the background, to perform various tasks. Generally, daemons start automatically at boot time and continue to run until shutdown or until you manually stop them. By convention, daemon names end with `d`.

2 Service Units Description

- A *systemd* unit is an abstract concept to define objects that the system knows how to manage.
- A unit has a name and a unit type
- The *systemd* daemon uses units to manage different types of objects:
 - **Service** units have a *.service* extension and represent system services. You can use service units to start often-accessed daemons, such as a web server.
 - **Socket** units have a *.socket* extension and represent inter-process communication (IPC) sockets that *systemd* should monitor. If a client connects to the socket, then the *systemd* manager starts a daemon and passes the connection to it. You can use socket units to delay the start of a service at boot time and to start less often used services on demand.
 - **Path** units have a extension and delay the activation of a service until a specific file-system change occurs. You can use *.path* path units for services that use spool directories, such as a printing system.
- To manage units, use the *systemctl* command.

3 List Service Units

```
[root@sads-lab ~]# systemctl list-units --type=service --all
UNIT                                LOAD    ACTIVE SUB    JOB    DESCRIPTION
accounts-daemon.service            loaded active running Accounts Service
alsa-restore.service               loaded inactive dead    Save/Restore Sound Card State
alsa-state.service                 loaded active running  Manage Sound Card State (restore
atd.service                        loaded active running  Deferred execution scheduler
auditd.service                     loaded active running  Security Auditing Service
● auto-cpufreq.service             not-found inactive dead    auto-cpufreq.service
● autofs.service                   not-found inactive dead    autofs.service
avahi-daemon.service               loaded active running  Avahi mDNS/DNS-SD Stack
blk-availability.service            loaded inactive dead    Availability of block devices
chronyd.service                    loaded active running  NTP client
colord.service                     loaded active running  Manage, Ins
cpupower.service                   loaded inactive dead    Configure C
crond.service                       loaded active running  Command Sch
cups.service                        loaded active running  CUPS Sched
dbus-broker.service                loaded active running  D-Bus System
dm-event.service                   loaded inactive dead    Device-mapp
dnf-makecache.service              loaded activating start start dnf makecach
```

`--type=service`
List units of type service

`--all`
List all units, even if they are not active. Omitting `--all` will only list active units.

Column	Description
UNIT	The service unit name.
LOAD	Whether the systemd daemon correctly parsed the unit's configuration and loaded the unit into memory.
ACTIVE	The high-level activation state of the unit. This information indicates whether the unit started successfully.
SUB	The low-level activation state of the unit. This information indicates more detailed information about the unit. The information varies based on unit type, state, and how the unit is executed.
DESCRIPTION	The short description of the unit.

3 List Service Units (Cont'd)

```
[root@sads-lab ~]# systemctl list-unit-files --type=service
```

UNIT FILE	STATE	PRESET
accounts-daemon.service	enabled	enabled
alsa-restore.service	static	-
alsa-state.service	static	-
arp-ethers.service	disabled	disabled
atd.service	enabled	enabled
auditd.service	enabled	enabled
autovt@.service	alias	-
avahi-daemon.service	enabled	enabled
blk-availability.service	disabled	disabled
bluetooth.service	enabled	enabled
bolt.service	static	-
brltty.service	disabled	disabled
canberra-system-bootup.service	disabled	disabled
canberra-system-shutdown-reboot.service	disabled	disabled
canberra-system-shutdown.service	disabled	disabled
chrony-wait.service	disabled	disabled
chronyd-restricted.service	disabled	disabled
chronyd.service	enabled	enabled

The ***systemctl*** command ***list-units*** option displays units that the ***systemd*** service attempts to parse and load into memory. This option does not display services that are installed but not enabled. You can use the ***systemctl*** command ***list-unit-files*** option to see the state of all the installed unit files

4 View Service States

```
[root@sads-lab ~]# systemctl status sshd.service
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena
   Active: active (running) since Sat 2024-05-11 21:12:36 +08; 23min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 914 (sshd)
     Tasks: 1 (limit: 10999)
    Memory: 1.8M
       CPU: 30ms
    CGroup: /system.slice/sshd.service
           └─914 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 11 21:12:36 sads-lab systemd[1]: Starting OpenSSH server daemon...
May 11 21:12:36 sads-lab sshd[914]: Server listening on 0.0.0.0 port 22.
May 11 21:12:36 sads-lab sshd[914]: Server listening on :: port 22.
May 11 21:12:36 sads-lab systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

Field	Description
Loaded	Whether the service unit is loaded into memory
Active	Whether the service unit is running, and if so, for how long
Docs	Where to find more information about the service
Main PID	The main process ID of the service, including the command name
Status	More information about the service
Process	More information about related processes
CGroup	More information about related control groups

4 View Service States (Cont'd)

Service States in the
Output of systemctl

Keyword	Description
loaded	The unit configuration file is processed.
active (running)	The service is running with continuing processes.
active (exited)	The service successfully completed a one-time configuration.
active (waiting)	The service is running but is waiting for an event.
inactive	The service is not running.
enabled	The service starts at boot time.
disabled	The service is not set to start at boot time.
static	The service cannot be enabled, but an enabled unit might start it automatically.
Keyword	Description

5 Verify the Status of a Service

Verify if a service unit is active (running)

systemctl is-active sshd.service

Verify if a service unit is enabled to start automatically during system boot

systemctl is-enabled sshd.service

```
[root@sads-lab ~]# systemctl is-active sshd.service
active
[root@sads-lab ~]# systemctl is-enabled sshd.service
enabled
[root@sads-lab ~]#
```

6 Start, stop, restart and reload a Service

To start a service

systemctl start sshd.service

To stop a service

systemctl stop sshd.service

```
[root@sads-lab ~]# systemctl stop sshd.service
[root@sads-lab ~]# systemctl is-active sshd.service
inactive
[root@sads-lab ~]# systemctl start sshd.service
[root@sads-lab ~]# systemctl is-active sshd.service
active
[root@sads-lab ~]#
```

When you restart a running service, the service first stops and then starts again. On the service restart, the new process gets a new ID during the startup and thus the process ID changes. To restart a running service, use the *systemctl* command *restart* option

```
[root@sads-lab ~]# systemctl status sshd.service | grep PID
Main PID: 3312 (sshd)
[root@sads-lab ~]# systemctl reload sshd.service
[root@sads-lab ~]# systemctl status sshd.service | grep PID
Process: 3354 ExecReload=/bin/kill -HUP $MAINPID (code=exited, status=0/SUCCESS)
Main PID: 3312 (sshd)
```

```
[root@sads-lab ~]# systemctl status sshd.service | grep PID
Main PID: 3285 (sshd)
[root@sads-lab ~]# systemctl restart sshd.service
[root@sads-lab ~]# systemctl status sshd.service | grep PID
Main PID: 3312 (sshd)
```

Some services can reload their configuration files without requiring a restart, which is called a *service reload*. Reloading a service does not change the process ID that is associated with various service processes. To reload a running service, use the *systemctl* command *reload* option

7 Enabling and Disabling a Service

Creating links in the *systemd* configuration directories enables the service to start at boot. You can create or remove these links by using the *systemctl* command with the *enable* or *disable* option.

To enable a service so that it automatically starts upon system reboot

```
# systemctl enable sshd.service
```

To disable a service so that it will not automatically restart upon system reboot

```
# systemctl disable sshd.service
```

```
[root@sads-lab ~]# systemctl status sshd.service | grep Loaded
    Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
[root@sads-lab ~]# systemctl disable sshd.service
Removed "/etc/systemd/system/multi-user.target.wants/sshd.service".
[root@sads-lab ~]# systemctl status sshd.service | grep Loaded
    Loaded: loaded (/usr/lib/systemd/system/sshd.service; disabled; preset: enabled)
[root@sads-lab ~]# systemctl enable sshd.service
Created symlink /etc/systemd/system/multi-user.target.wants/sshd.service → /usr/lib/systemd/system/sshd.service.
[root@sads-lab ~]# systemctl status sshd.service | grep Loaded
    Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
```

8 Summary of systemctl commands to manage units

Keyword	Description
view detailed information about a unit state	systemctl status UNIT
stop a service on a running system	systemctl stop UNIT
start a service on a running system	systemctl start UNIT
restart a service on a running system	systemctl restart UNIT
reload configuration file of a running service	systemctl reload UNIT
configure a service to start at boot time	systemctl enable UNIT
disable a service from starting at boot time	systemctl disable UNIT
list units which are required by the specified unit	systemctl list-dependencies UNIT

9 Introduction to Network Connection

- Being “connected” is important for a Linux server to achieve many of its objectives. Typically, many software packages that can be installed on a Linux machine is usually obtained from the Internet (although software package installation from media is still possible, it is a rarity nowadays). Many of the services such as web, domain name, virtual private networks only makes sense when the Linux machine is connected to the Internet.
- Configuring the Linux machine for connectivity is the responsibility of the root user on the system. Unfortunately, there are many variables involved in network configuring that makes the task complex and requires a deep understanding of network protocols and terms such as NAT, bridging, host-only adapter, virtual adapters etc.
- There are also many different tasks involved in the network configuration that involves domain name services, default gateways, routing and dynamic host configuration protocols.
- This module will focus on how to configure your system to connect to the network and troubleshoot networking issues.

10 Retrieving Network Interface Settings

- The term network interface refers to a point of connection between a computer and a network. It can be implemented in either hardware or software. The network interface card (NIC) is an example of the hardware interface while the loopback interface (127.0.0.1) is an example of the software interface.
- The `ip address` command is used to retrieve the current network interface settings.

```
[root@sads-lab ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a4:91:98 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86357sec preferred_lft 86357sec
    inet6 fe80::a00:27ff:fea4:9198/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:fc:3d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::db97:c47f:b85e:2ced/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Network
Interface
labels

IP
addresses
+ CIDR

Compare the `ip addr` command output to the Network settings for the VM in Virtualbox. Interface `enp0s3` is Adapter 1 and is connected to the Internet (NAT)

Adapter 1 Adapter 2 Adapter 3 Adapter 4

☒ Enable Network Adapter

Attached to: NAT

Name:

Advanced

Adapter Type: Intel PRO/1000 MT Desktop (82540EM)

Promiscuous Mode: Deny

MAC Address: 080027A49198

☒ Cable Connected

Port Forwarding

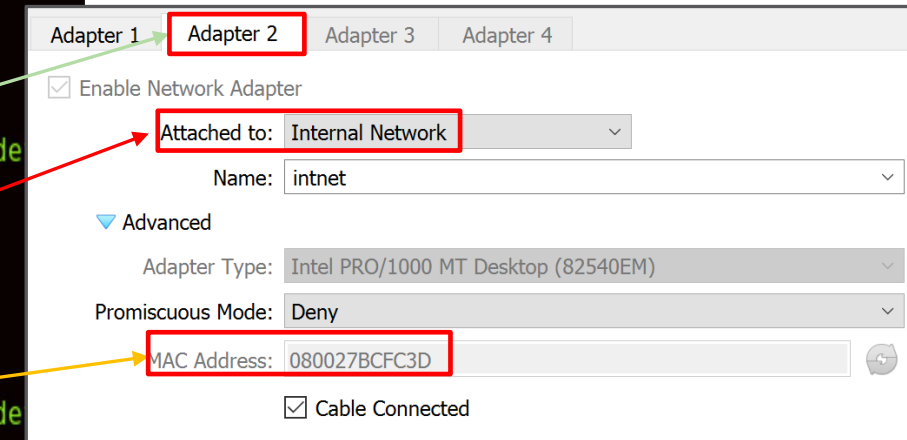
10 Retrieving Network Interface Settings (Cont'd)

```
[root@sads-lab ~]# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a4:91:98 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 86357sec preferred_lft 86357sec
    inet6 fe80::a00:27ff:fea4:9198/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:fc:3d brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::db97:c47f:b85e:2ced/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Network
Interface
labels

IP
addresses
+ CIDR

Compare the ip addr command output to the Network settings for the VM in Virtualbox. Interface enp0s8 is Adapter 2 and is connected to Internal Network named intnet.



11 Retrieving and modifying the IP routing table

- You can retrieve the routing table in the linux system by executing the command `ip route`

```
[root@sads-lab ~]# ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.100 metric 101
```

- Notice the first line of the output indicating that the default route is via `10.0.2.2` which is outgoing from the `enp0s3` network interface (Adapter 1 NAT)
- You can add and delete routes by using the `route` command. The example shows the addition of a new static route towards the `172.16.0.0/16` network via the `enp0s8` network interface.

```
[root@sads-lab ~]# route add -net 172.16.0.0 netmask 255.255.0.0 dev enp0s8
[root@sads-lab ~]# ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
172.16.0.0/16 dev enp0s8 scope link
192.168.1.0/24 dev enp0s8 proto kernel scope link src 192.168.1.100 metric 101
```

12 Displaying Network Ports and Services

- TCP services use sockets as endpoints for communication, and are composed of an IP address, protocol, and port number. Services typically listen on standard ports, whereas clients use a random available port. Well-known names for standard ports are listed in the /etc/services file.
- The ss command is used to display socket statistics. The ss command replaces the earlier netstat tool, from the net-tools package, which might be more familiar to some system administrators but is not always installed.

```
[root@sads-lab ~]# ss -ta
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process
LISTEN     0          128      0.0.0.0:ssh           0.0.0.0:*
LISTEN     0          4096     127.0.0.1:ipp         0.0.0.0:*
LISTEN     0          4096     [::1]:ipp            [::]:*
LISTEN     0          128      [::]:ssh              [::]:*
```

0.0.0.0:ssh or *:ssh means that the port that is used for SSH is listening on all IPv4 addresses

Option	Description
-n	Show numbers instead of names for interfaces and ports.
-t	Show TCP sockets.
-u	Show UDP sockets.
-l	Show only listening sockets.
-a	Show all (listening and established) sockets.
-p	Show the process that uses the sockets.
-A inet	Display active connections (but not listening sockets) for the inet address family. That is, ignore local UNIX domain sockets. For the ss command, both IPv4 and IPv6 connections are displayed. For the netstat command, only IPv4 connections are displayed. (The netstat -A inet6 command displays IPv6 connections, and the netstat -46 command displays IPv4 and IPv6 at the same time.)

Options for ss and netstat

13 Introduction to the NetworkManager Service

- The NetworkManager service monitors and manages a system's network settings. Service configuration files are stored in the `/etc/NetworkManager/system-connections/` directory.
- The NetworkManager service manages network devices and connections. A device is a physical or virtual network interface that provides for network traffic. A connection has related configuration settings for a single network device. Each connection must have a unique name or ID, which may often (by default) match the device name that it configures.
- A single device can have multiple connection configurations and switch between them, but only one connection can be active per device.

14 The nmcli command

- Use the *nmcli* utility to create and edit connection files from the command line. The *nmcli device status* command displays the status of all network devices:

```
[root@sads-lab ~]# nmcli dev status
DEVICE  TYPE      STATE              CONNECTION
enp0s3  ethernet  connected          enp0s3
enp0s8  ethernet  connected          lab-vlan
lo      loopback  connected (externally) lo
```

- The *nmcli connection show* command displays a list of all connections. Use the *--active* option to list only active connections.

```
[root@sads-lab ~]# nmcli con show
NAME      UUID                                  TYPE      DEVICE
enp0s3    4fb23777-1553-37ac-b779-af6116990386 ethernet  enp0s3
lo        4b9e3e50-c19d-4d6f-9de1-dacf923dbbbf loopback   lo
lab-vlan  fcfa4196-0ebf-4445-9c65-85a444cd8414 ethernet  --
[root@sads-lab ~]# nmcli con show --active
NAME      UUID                                  TYPE      DEVICE
enp0s3    4fb23777-1553-37ac-b779-af6116990386 ethernet  enp0s3
lo        4b9e3e50-c19d-4d6f-9de1-dacf923dbbbf loopback   lo
```

lab-vlan
connection is
inactive

14 The nmcli command (cont'd)

- Use the *nmcli connection add* command to add network connections. The data for the added network connection is stored in the */etc/NetworkManager/system-connections/* directory as a file with a *.nmconnection* suffix.

```
[root@sads-lab ~]# nmcli con add con-name Internet ifname enp0s3 type ethernet
Connection 'Internet' (8a366f36-a145-40fd-93ad-f2d2f6d4050a) successfully added.
[root@sads-lab ~]# ls /etc/NetworkManager/system-connections/
enp0s3.nmconnection  Internet.nmconnection  lab-vlan.nmconnection
```

- The next example creates an *static* connection of the *ethernet* type for the *static* network interface with a static IPv4 network setting. This command configures the 192.168.2.100 IP address with a network prefix of /24 and a network gateway of 192.168.2.1. The *nmcli* connection add command fails if the connection name that you try to add exists.

```
[root@sads-lab ~]# nmcli con add con-name static ifname enp0s8 type ethernet ipv4.method manual
l ipv4.address 192.168.2.100/24 ipv4.gateway 192.168.2.1
Connection 'static' (9c8525de-4b33-4e6e-bec9-fbeac035321e) successfully added.
[root@sads-lab ~]# ls /etc/NetworkManager/system-connections/
enp0s3.nmconnection  Internet.nmconnection  lab-vlan.nmconnection  static.nmconnection
```

14 The nmcli command (cont'd)

- Use the following command to update the *static* connection to set the *192.0.2.2/24* IPv4 address and the *192.0.2.254* default gateway. Use the *nmcli* command *connection.autoconnect* parameter to automatically enable or disable the connection at system boot.

```
[root@sads-lab ~]# ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:fc:3d brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.100/24 brd 192.168.2.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a915:432b:83ad:6f32/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@sads-lab ~]# nmcli con show --active
NAME      UUID                                  TYPE      DEVICE
enp0s3    4fb23777-1553-37ac-b779-af6116990386 ethernet  enp0s3
static    20de8753-3e27-4ee8-b999-9c641bf16fbd ethernet  enp0s8
lo        4022a7d7-e2e2-4b67-92a9-6fd015391f93 loopback   lo
[root@sads-lab ~]# nmcli con mod static ipv4.addresses 192.0.2.2/24 ipv4.gateway 192.0.2.254 autoconnect yes
[root@sads-lab ~]# nmcli con up static
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
[root@sads-lab ~]# ip addr show enp0s8
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:bc:fc:3d brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.2/24 brd 192.0.2.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a915:432b:83ad:6f32/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

After the connection modify command, you need to issue the *nmcli connection up <connection name>* command to make the modification effective.

15 Edit Network Configuration Files

- An alternative to managing and configuring network connections using the *nmcli* command, you can also edit the network configuration files to achieve the same objectives. All network connection configuration files are stored in the `/etc/NetworkManager/system-connections` folder.

```
[root@sads-lab ~]# ls /etc/NetworkManager/system-connections/  
enp0s3.nmconnection  Internet.nmconnection  lab-vlan.nmconnection  static.nmconnection
```

- The following table summarises the key sections in the file and the parameters (to be continued next page):

nmcli con mod	*.nmconnection file	Effect
ipv4.method manual	[ipv4] method=manual	Configure IPv4 addresses statically.
ipv4.method auto	[ipv4] method=auto	Look for configuration settings from a DHCPv4 server. It shows static addresses only when it has information from DHCPv4.
ipv4.addresses 192.0.2.1/24	[ipv4] address1=192.0.2.1/24	Set a static IPv4 address and network prefix. For more than one connection address, the address2 key defines the second address, and the address3 key defines the third address.
ipv4.gateway 192.0.2.254	[ipv4] gateway=192.0.2.254	Set the default gateway.
ipv4.dns 8.8.8.8	[ipv4] dns=8.8.8.8	Modify <code>/etc/resolv.conf</code> to use this name server.
ipv4.dns-search example.com	[ipv4] dns-search=example.com	Modify <code>/etc/resolv.conf</code> to use this domain in the search directive.
ipv4.ignore-auto-dns true	[ipv4] ignore-auto-dns=true	Ignore DNS server information from the DHCP server.

15 Edit Network Configuration Files (Cont'd)

nmcli con mod	*.nmconnection file	Effect
connection.autoconnect yes	[connection] autoconnect=true	Automatically activate this connection at boot.
connection.id ens3	[connection] id=Main eth0	The name of this connection.
connection.interface-name ens3	[connection] interface-name=ens3	The connection is bound to the network interface with this name.
802-3-ethernet.mac-address ...	[802-3-ethernet] mac-address=	The connection is bound to the network interface with this MAC address.

After you modify the configuration files, run the *nmcli con reload* command so that *NetworkManager* loads the configuration changes. If the *autoconnect* variable is set to *false* in the connection profile, then activate the connection manually. The screen capture on the right shows the content of the *static.nmconnection*

```
[connection]
id=static
uuid=20de8753-3e27-4ee8-b999-9c641bf16fbd
type=ethernet
interface-name=enp0s8
timestamp=1716007355

[ethernet]

[ipv4]
address1=192.0.2.2/24,192.0.2.254
method=manual

[ipv6]
addr-gen-mode=default
method=auto
```

16 Setting the Hostname

- The hostname is used to identify the system by applications such as web servers. On both Debian-derived and Red Hat-derived systems, the `/etc/hostname` file contains this information. This file is read at boot time to set the host name.
- The `hostnamectl` command is used to set and view the system's host and domain name. It is the system administrator's responsibility to assign an appropriate hostname. It cannot be longer than 64 characters and can contain alphanumeric characters, periods and dashes only.
- Execute the `hostnamectl` command with the `set-hostname` command followed by the fully qualified domain name to set the hostname correctly.

```
[root@sads-lab ~]# hostnamectl set-hostname centos.sads.com
[root@sads-lab ~]# hostname -f
centos.sads.com
[root@sads-lab ~]# hostname -s
centos
[root@sads-lab ~]# hostname -d
sads.com
```

```
[root@sads-lab ~]# hostnamectl
Static hostname: centos.sads.com
Icon name: computer-vm
Chassis: vm 01F5B4
Machine ID: 345e9f1b658c4ad6a26032a469a73dec
Boot ID: 72b03ce6a0034a4da33dafa0315f3a49
Virtualization: oracle
Operating System: CentOS Stream 9
CPE OS Name: cpe:/o:centos:centos:9
Kernel: Linux 5.14.0-435.el9.x86_64
Architecture: x86-64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Firmware Version: VirtualBox
```

16 Resolving Hostnames to IP address

- Order of hostname resolution providers is configured in the `/etc/nsswitch.conf` file. Typically the order of resolution is to first resolve using the `/etc/hosts` file and if not found, then use Domain Name Servers specified in `/etc/resolv.conf` file

```
[root@sads-lab ~]# cat /etc/hosts
```

```
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
```

```
:::1        localhost localhost.localdomain localhost6 localhost6.localdomain6
```

```
[root@sads-lab ~]# cat /etc/resolv.conf
```

```
# Generated by NetworkManager
```

```
search sads.com
```

```
nameserver 8.8.8.8
```

- search* : A list of domain names to try with a short hostname. Either *search* or *domain* should be set in the same file; if they are both set, then only the last entry takes effect.
- nameserver* : The IP address of a name server to query. Up to three name server directives can be given to provide backups if one name server is down.

```
[root@sads-lab ~]# nmcli con mod static ipv4.dns 192.168.2.1
```

```
[root@sads-lab ~]# nmcli con mod static +ipv4.dns 4.4.2.2
```

```
[root@sads-lab ~]# nmcli con down static
```

```
Connection 'static' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/5)
```

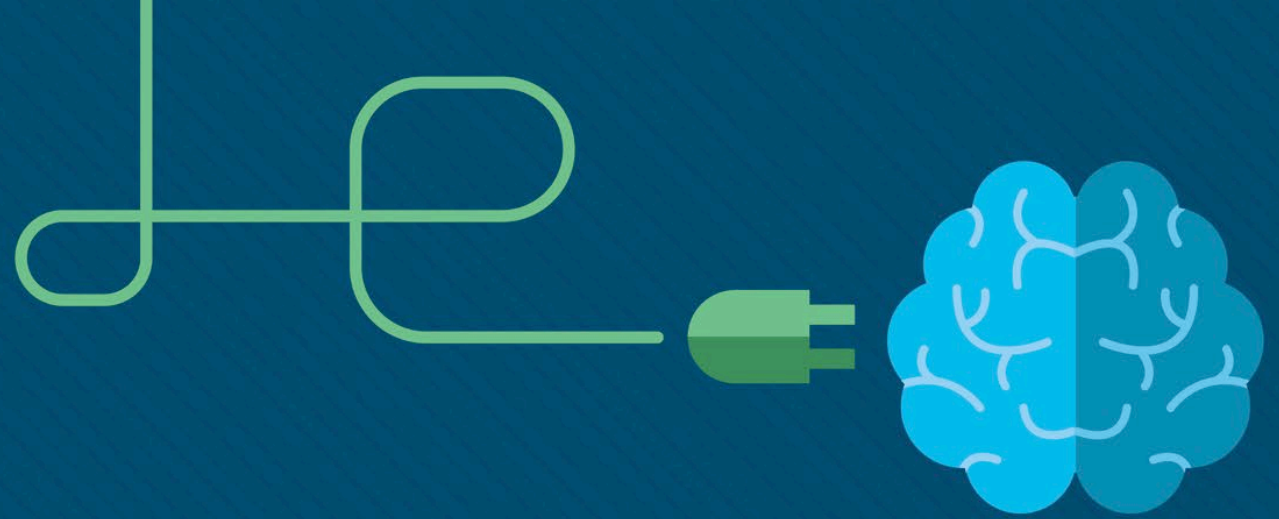
```
[root@sads-lab ~]# nmcli con up static
```

```
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
```

```
[root@sads-lab ~]# cat /etc/NetworkManager/system-connections/static.nmconnection
```

```
.....  
[ipv4]  
address1=192.0.2.2/24,192.0.2.254  
dns=192.168.2.1;4.4.2.2;  
method=manual  
.....
```

```
[root@sads-lab ~]# cat /etc/resolv.conf  
# Generated by NetworkManager  
search sads.com  
nameserver 8.8.8.8  
nameserver 192.168.2.1  
nameserver 4.4.2.2
```



Lab

RH124 Chapter 9.5 Lab Control Services and Daemons

RH124 Chapter 11.11 Lab Managing Network



Practice Exercises

1 Managing Services with systemctl

- List all active services on the system.
- Check if the sshd service is running.
- Start and enable the firewalld service to run at boot.
- Restart crond and check its status.
- Stop the cups service and disable it from starting at boot.

2 Networking Basics & Configuration

- Display all network interfaces and their assigned IP addresses.
- Show the system's default gateway.
- Use ss to list all listening TCP ports.
- Retrieve the hostname and change it to student-PC.
- Verify that the hostname change was applied.

3 Network Manager & nmcli

- List all active network connections.
- Display detailed information about a specific connection.
- Add a new static network connection with IP 192.168.2.100/24.
- Modify an existing connection to use 8.8.8.8 as the primary DNS server.
- Bring the modified connection up.